

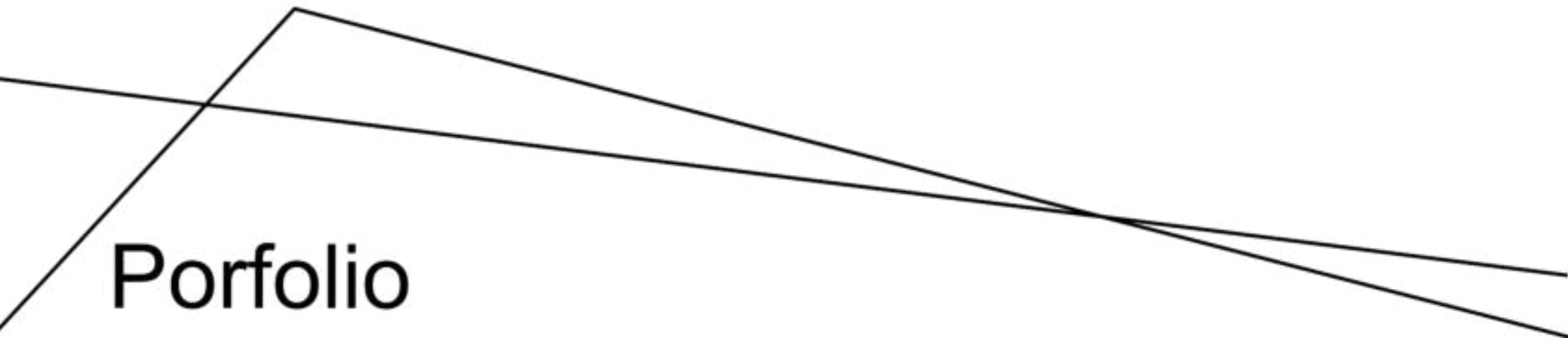


Portfolio

Robin s. Jesensky

ARCHIP Year 1 – Semester 2

Rethinking Logistics



Portfolio

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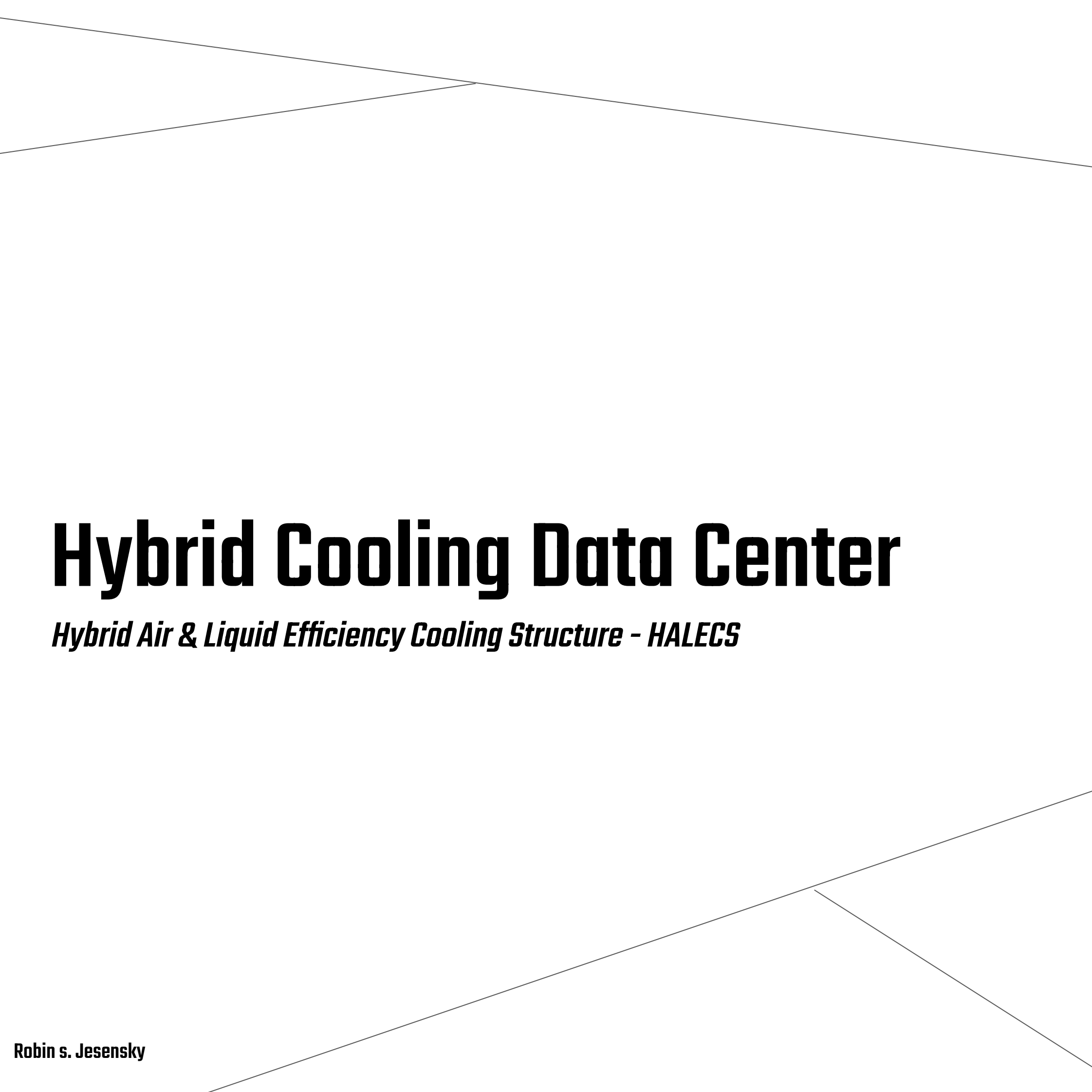
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The background features several thin, dark gray lines that intersect to form abstract, angular shapes. One line runs diagonally from the top left towards the center. Another line runs diagonally from the top right towards the center. A third line runs diagonally from the bottom left towards the center. A fourth line runs diagonally from the bottom right towards the center. These lines create a sense of depth and architectural structure.

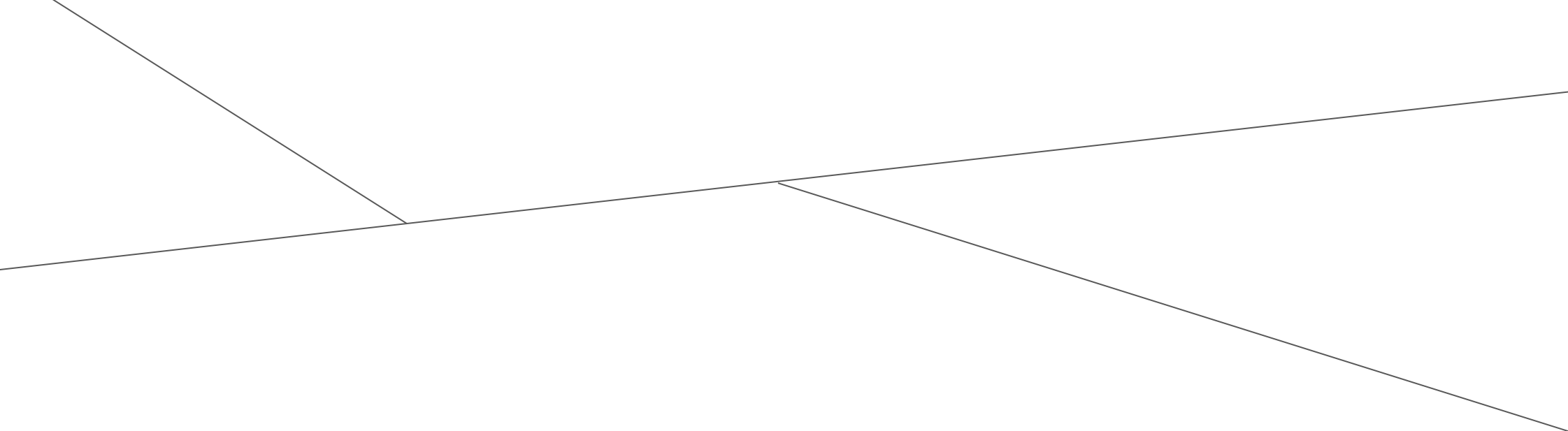
Hybrid Cooling Data Center

Hybrid Air & Liquid Efficiency Cooling Structure - HALECS

The Problem

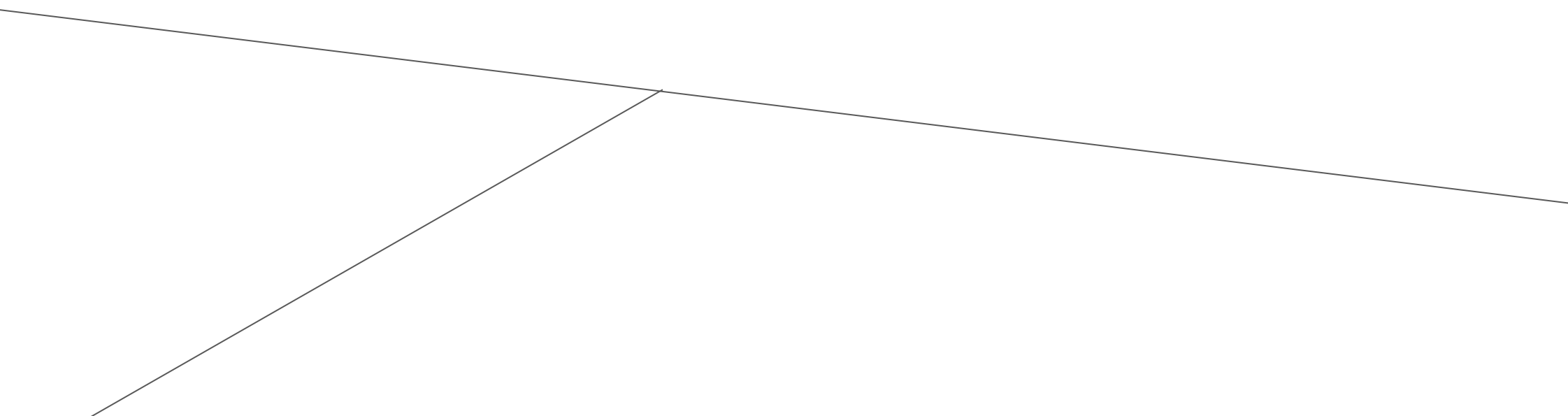
Data Centers:

- Use too much water
- Use too much electricity (on cooling)
- Are inefficient space-wise



The Solution

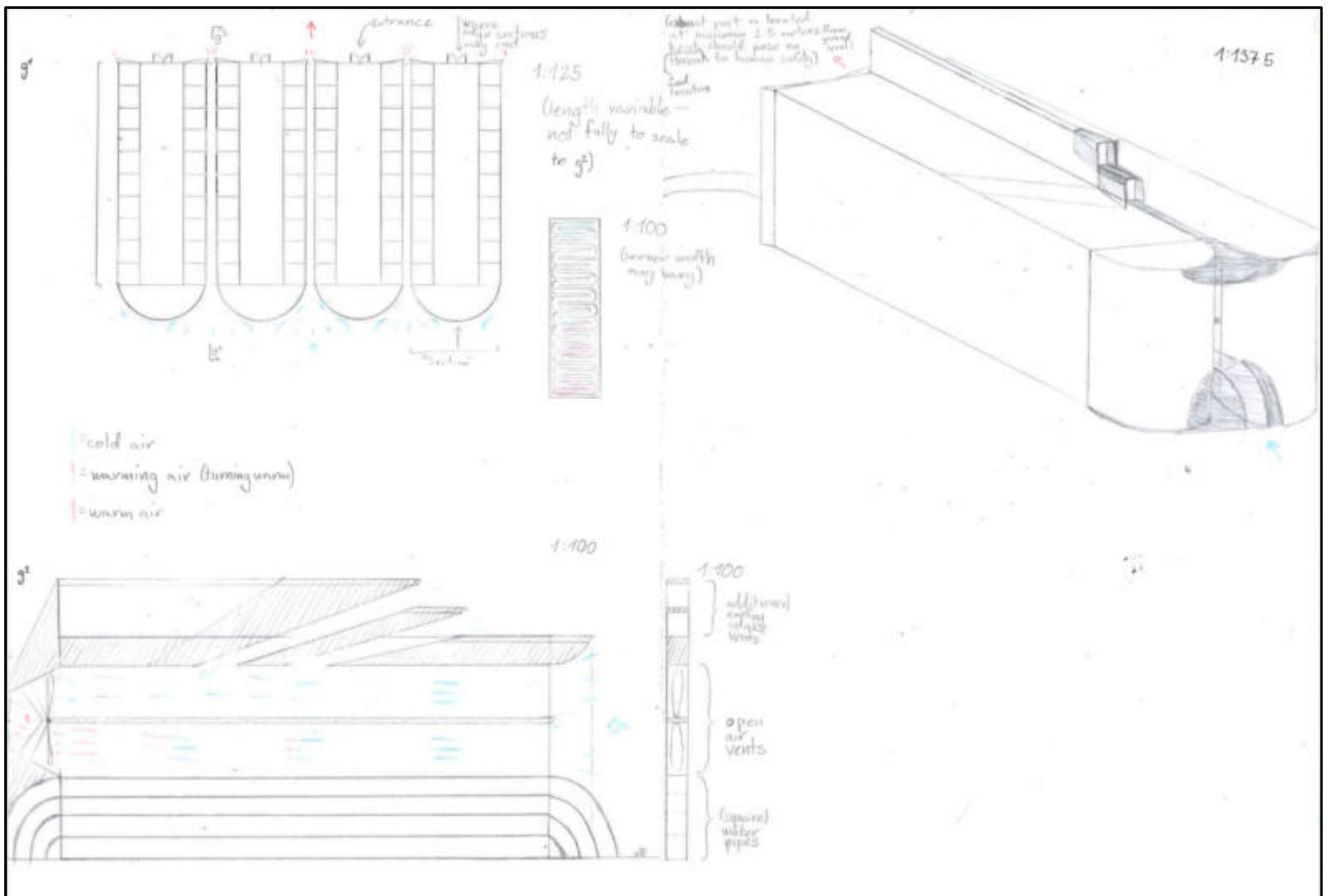
Data Centers:

- Lessen water use
 - Lessen electricity use (on cooling at least)
 - Utilize space more efficiently
- 

The Technology

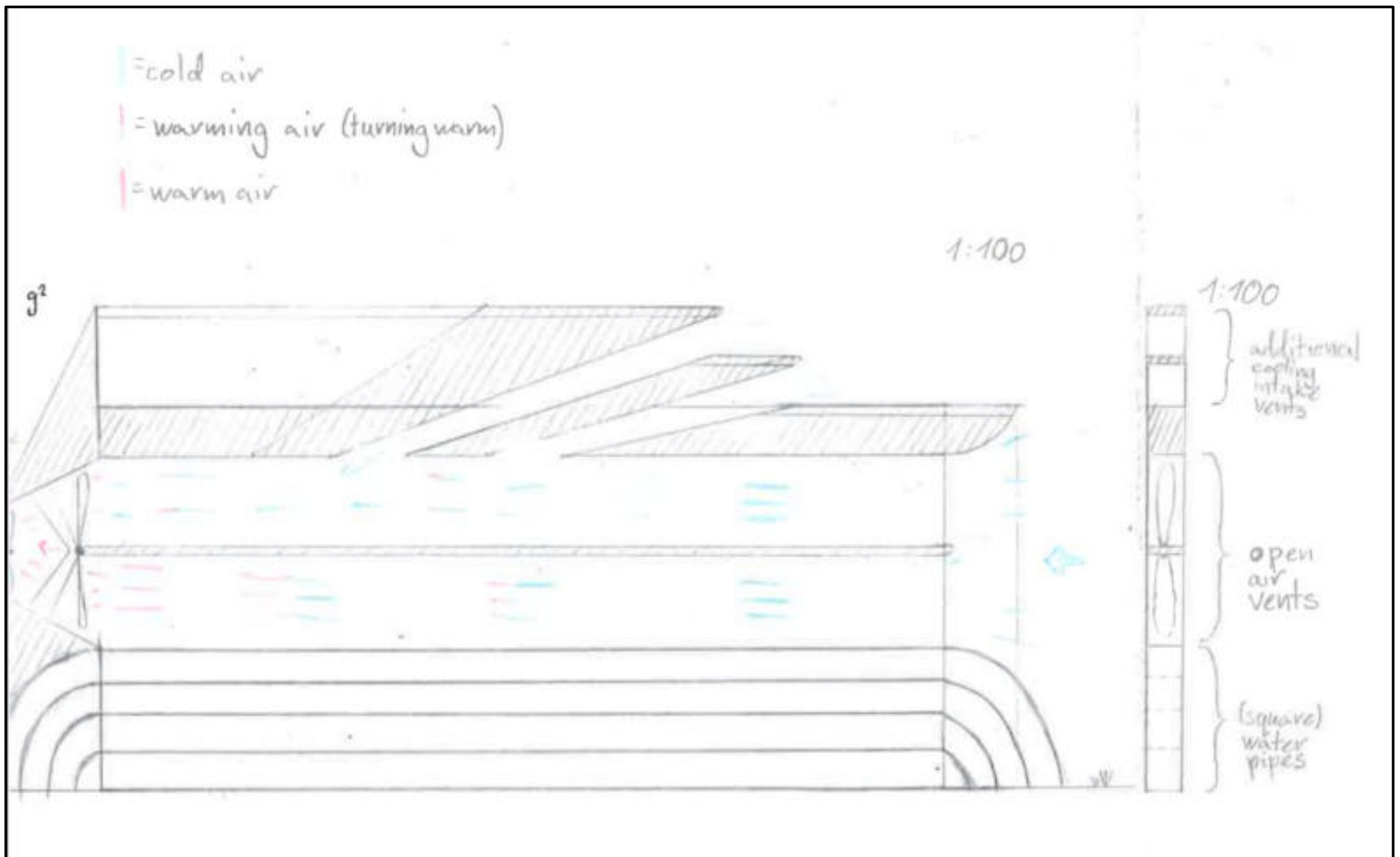
The Technology

First Concept Sketch



The Technology

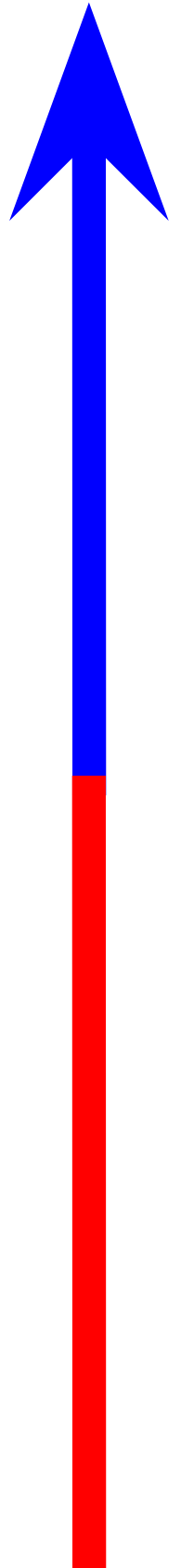
First Concept Sketch



The Technology

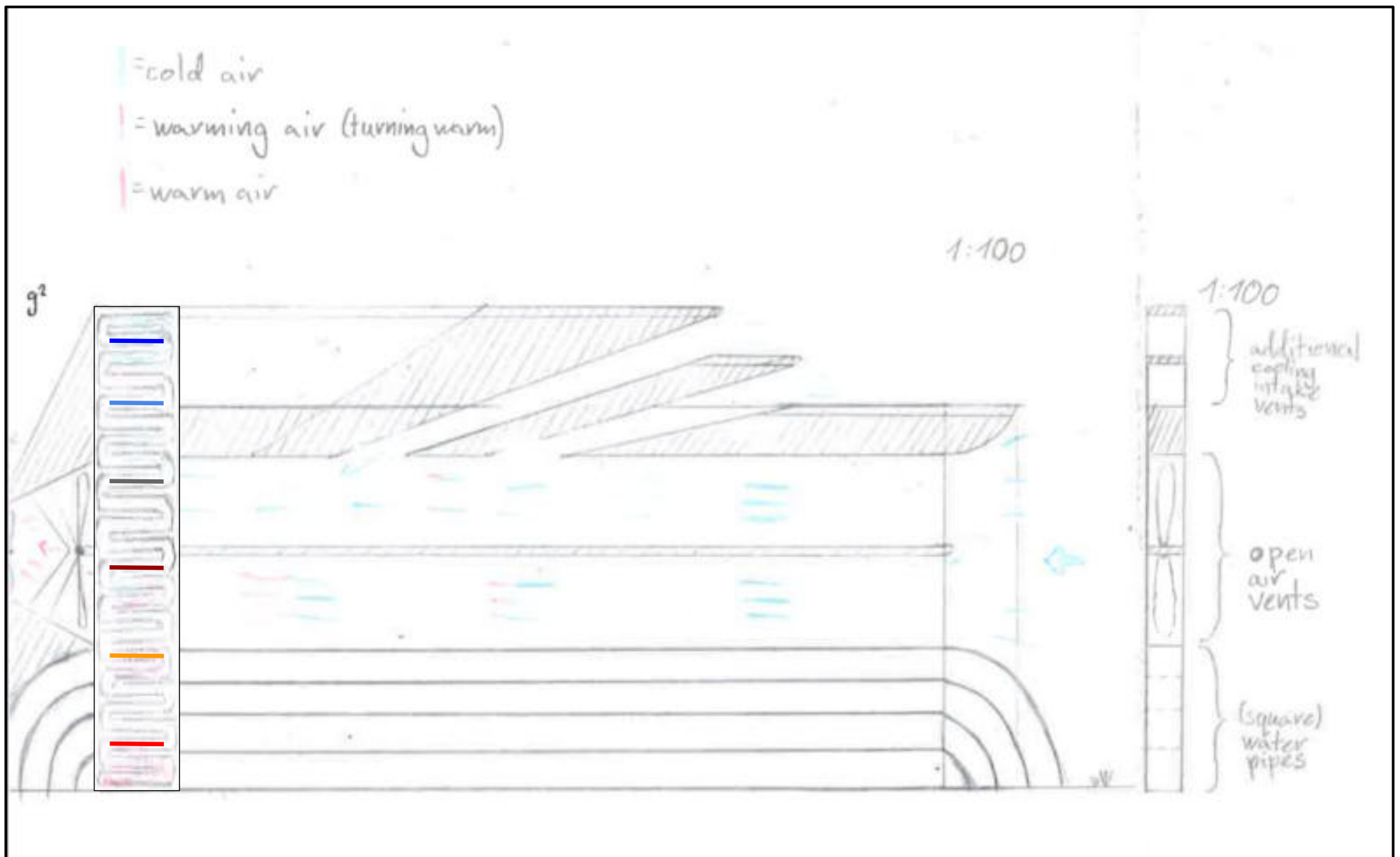
First Concept Sketch

-Individual Server Cooling Loop-



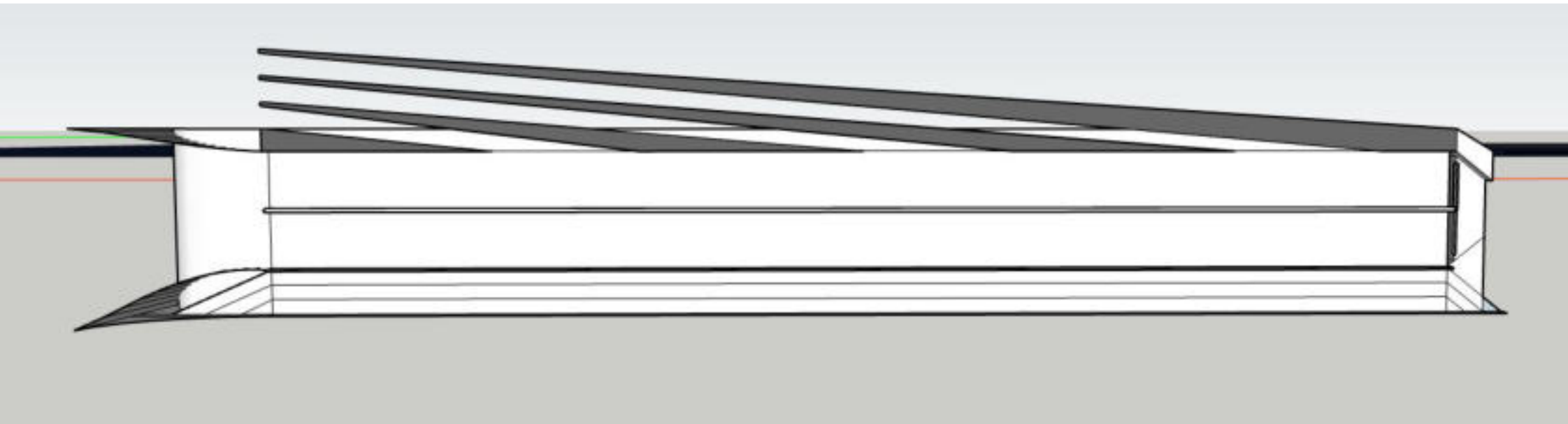
The Technology

First Concept Sketch



The Technology

Model V1 -Section1-



The Technology

Model V1 -Section1-



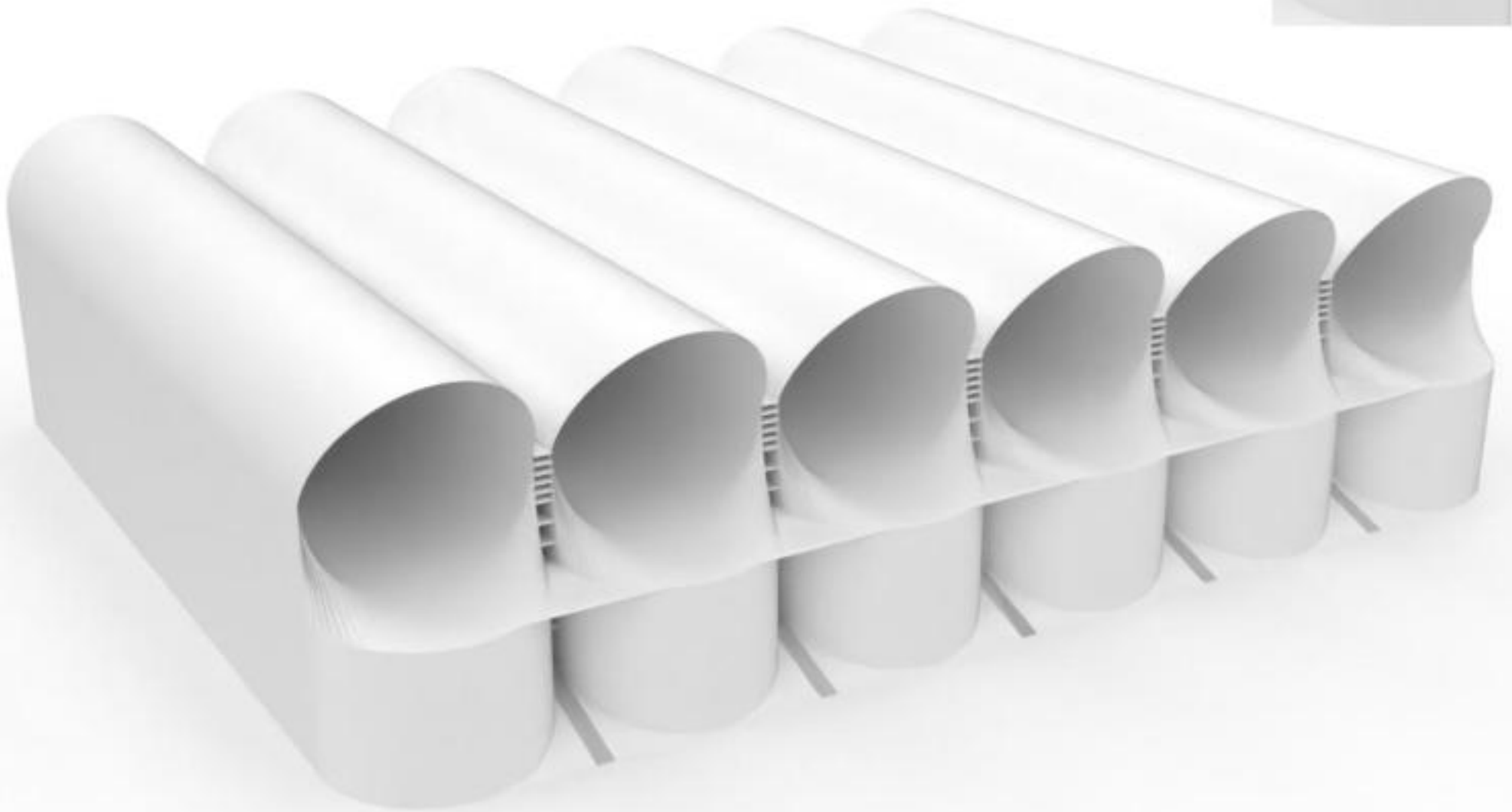
The Technology

Model V6 -Terra Visualization-



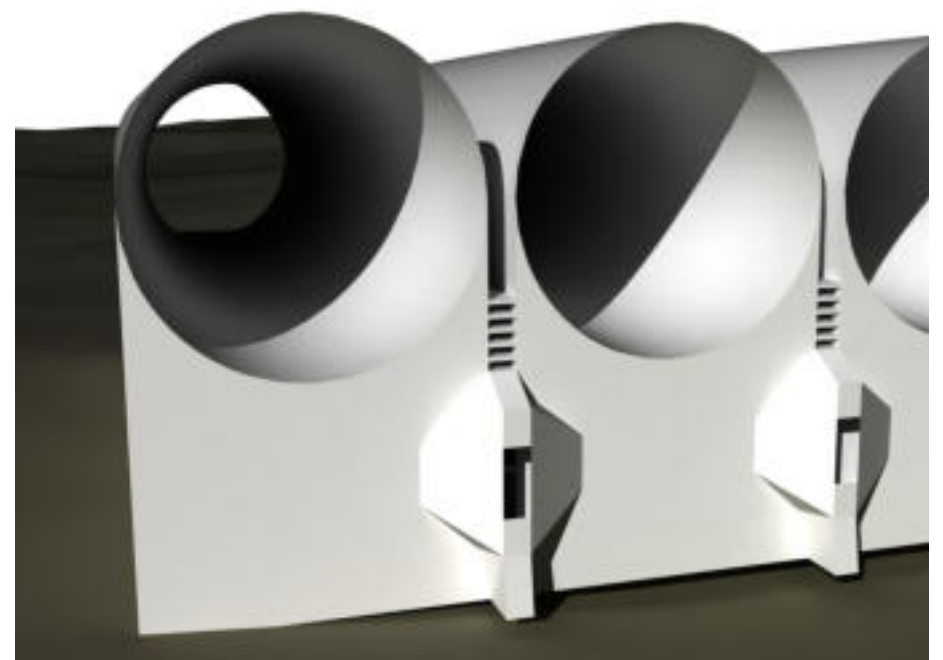
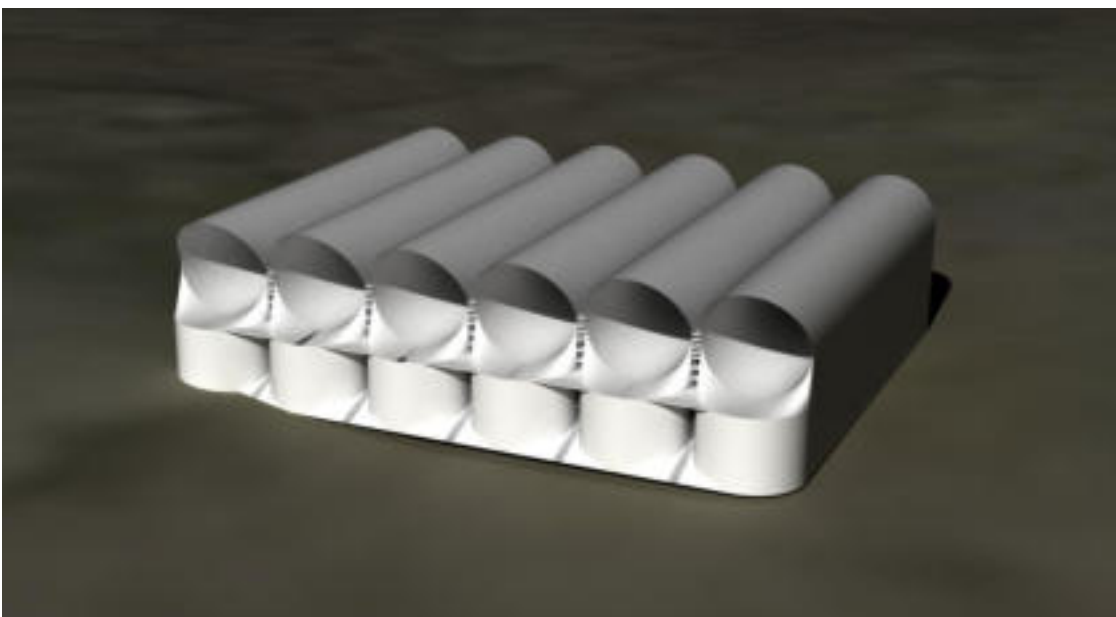
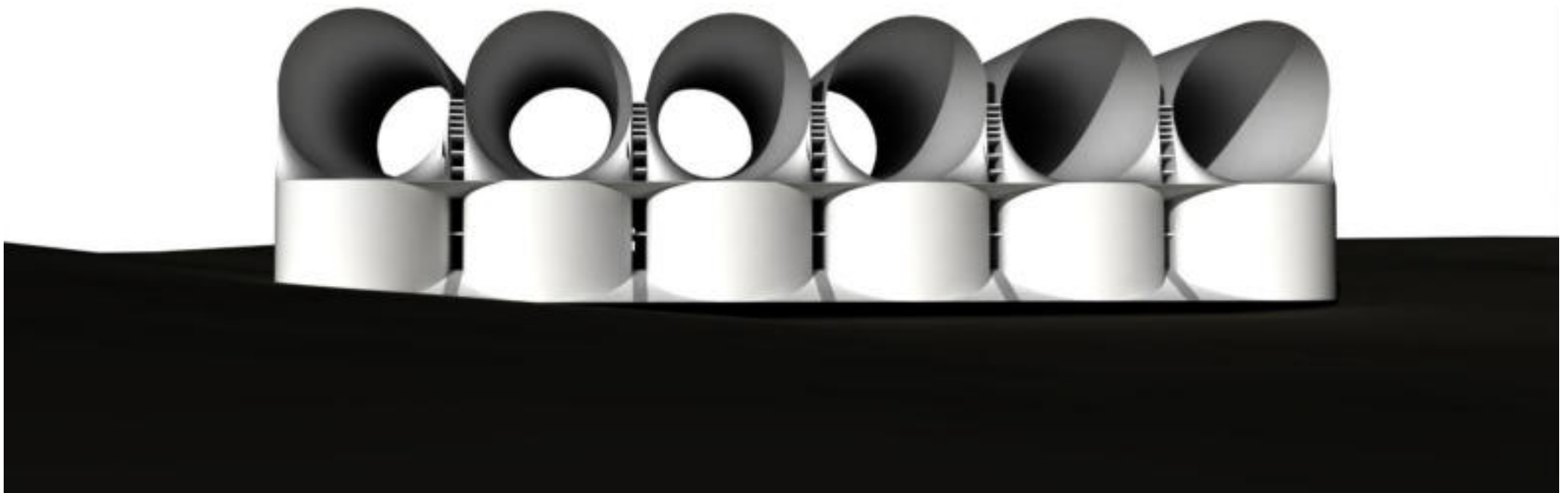
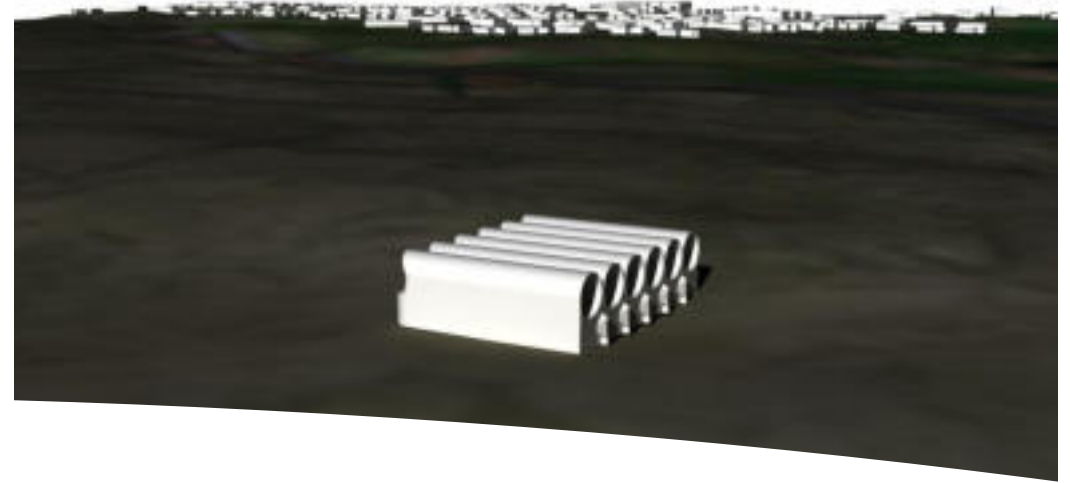
The Technology

Model V11 -Clean Visualization-



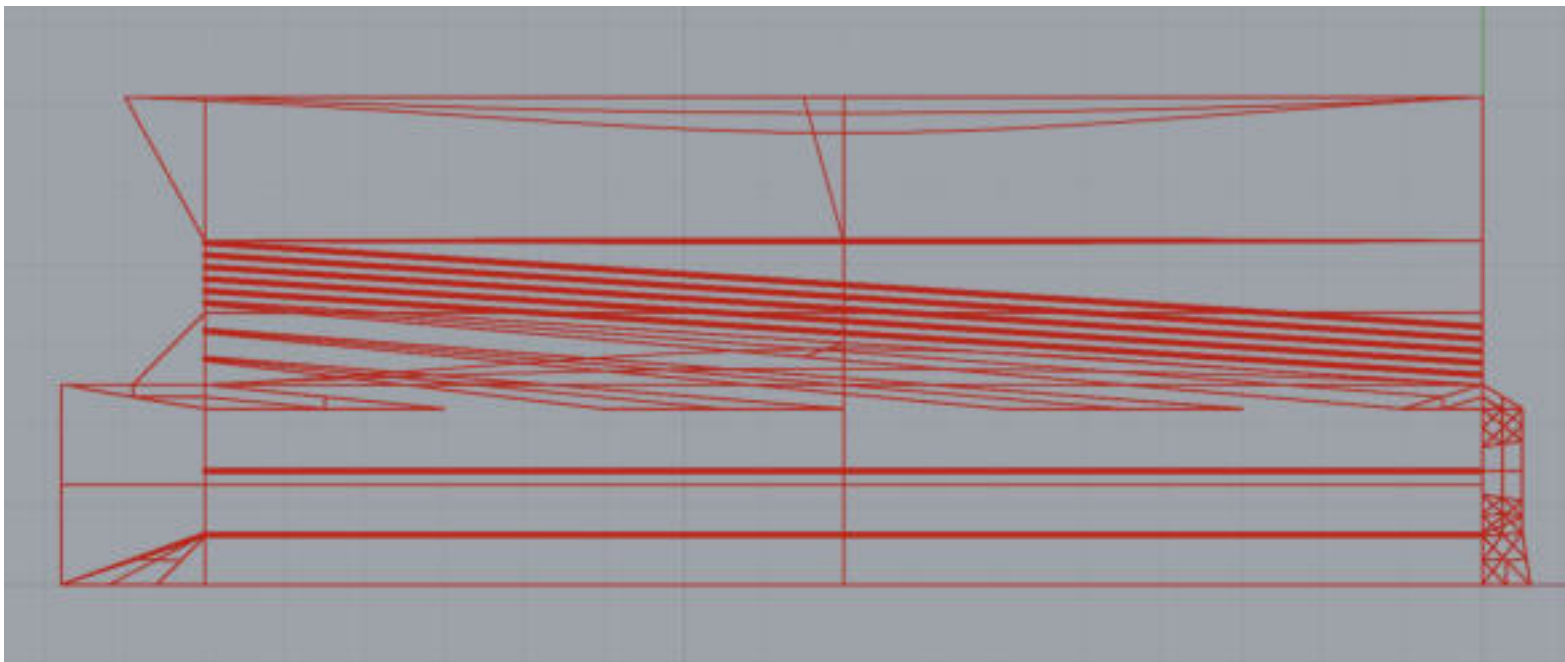
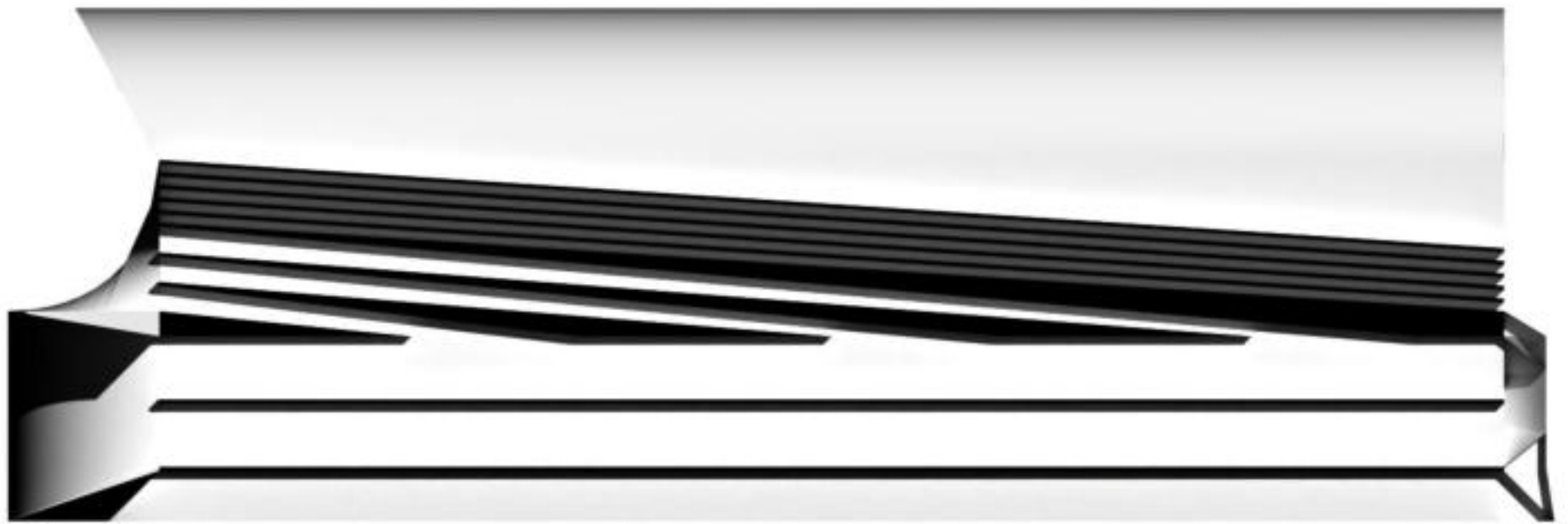
The Technology

Model V11 -Terra Visualization-



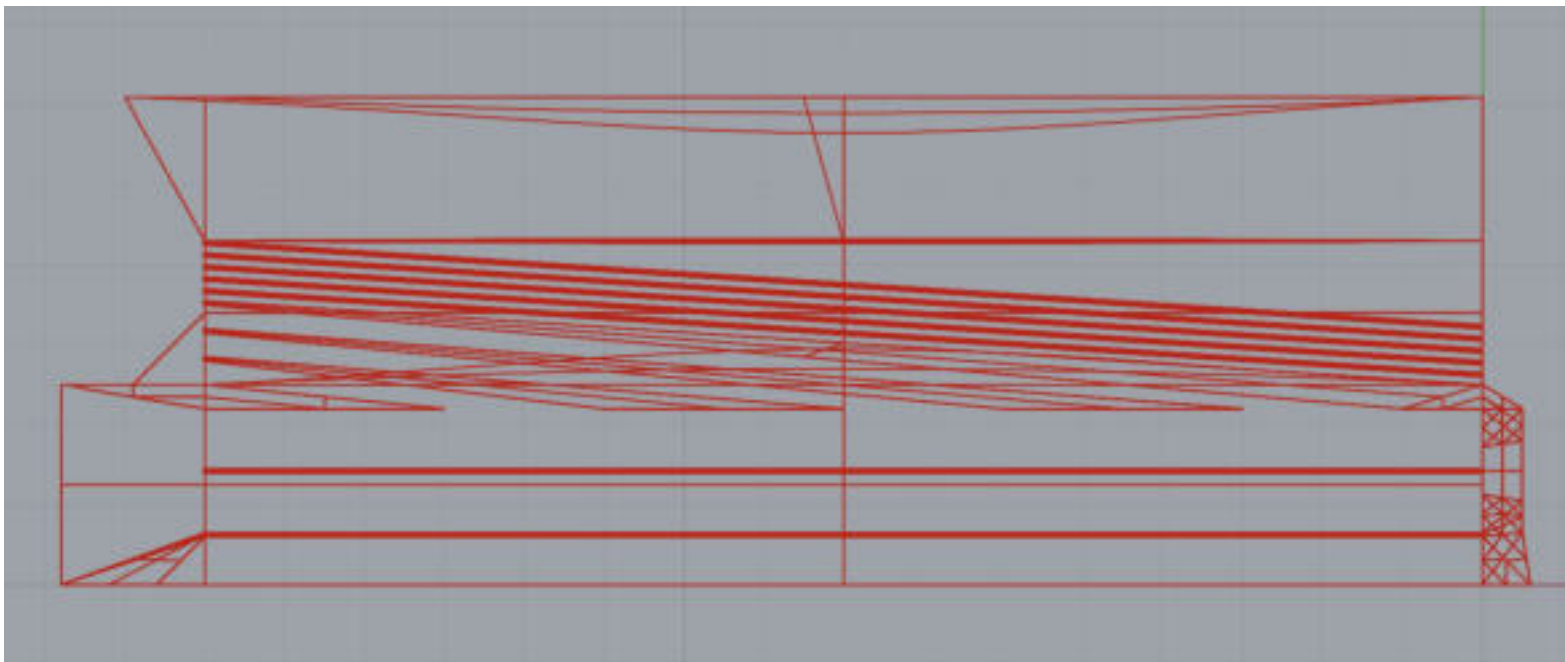
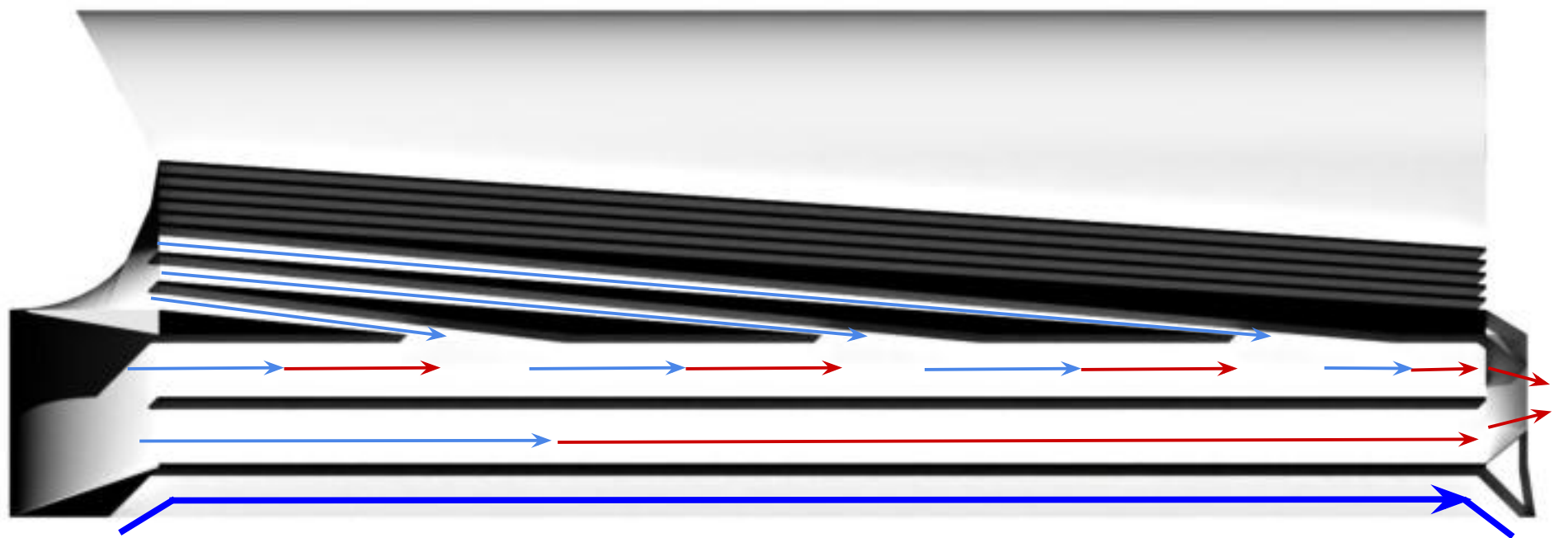
The Technology

Model V11 -Section-



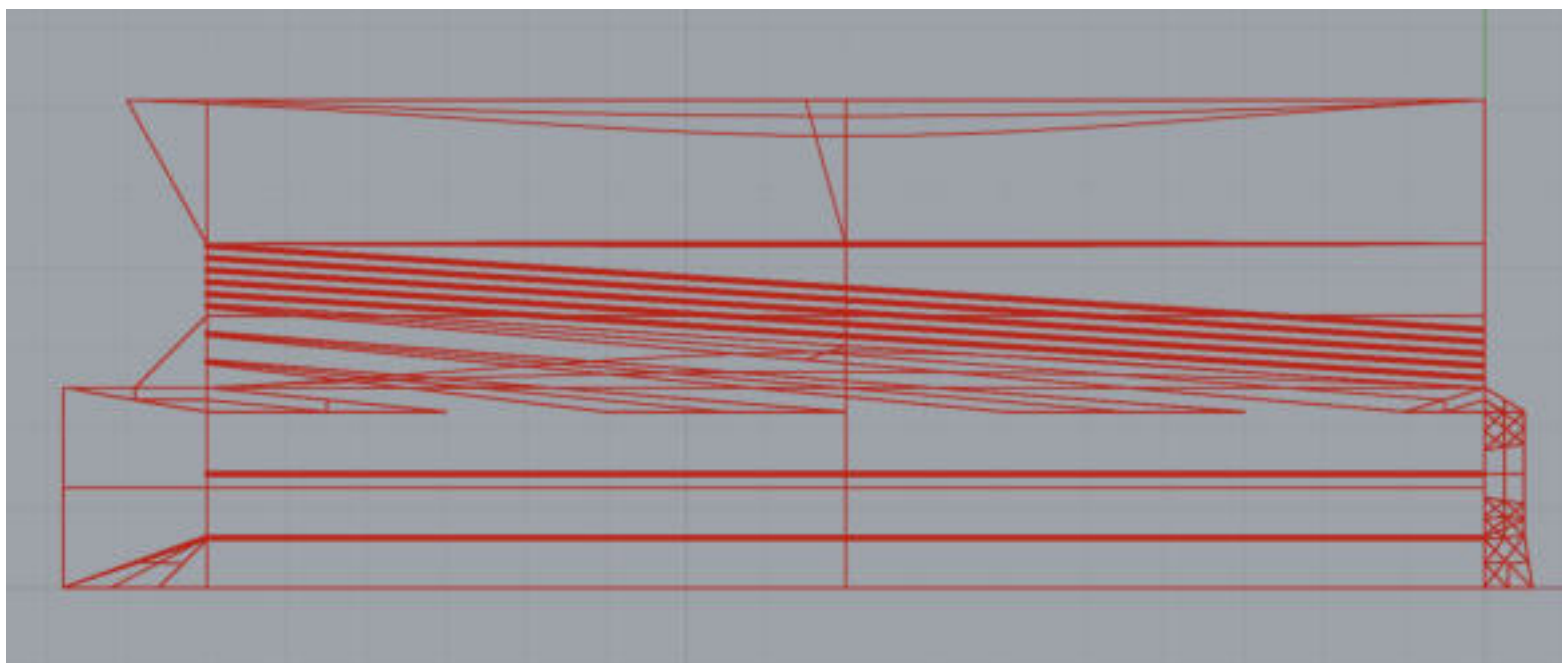
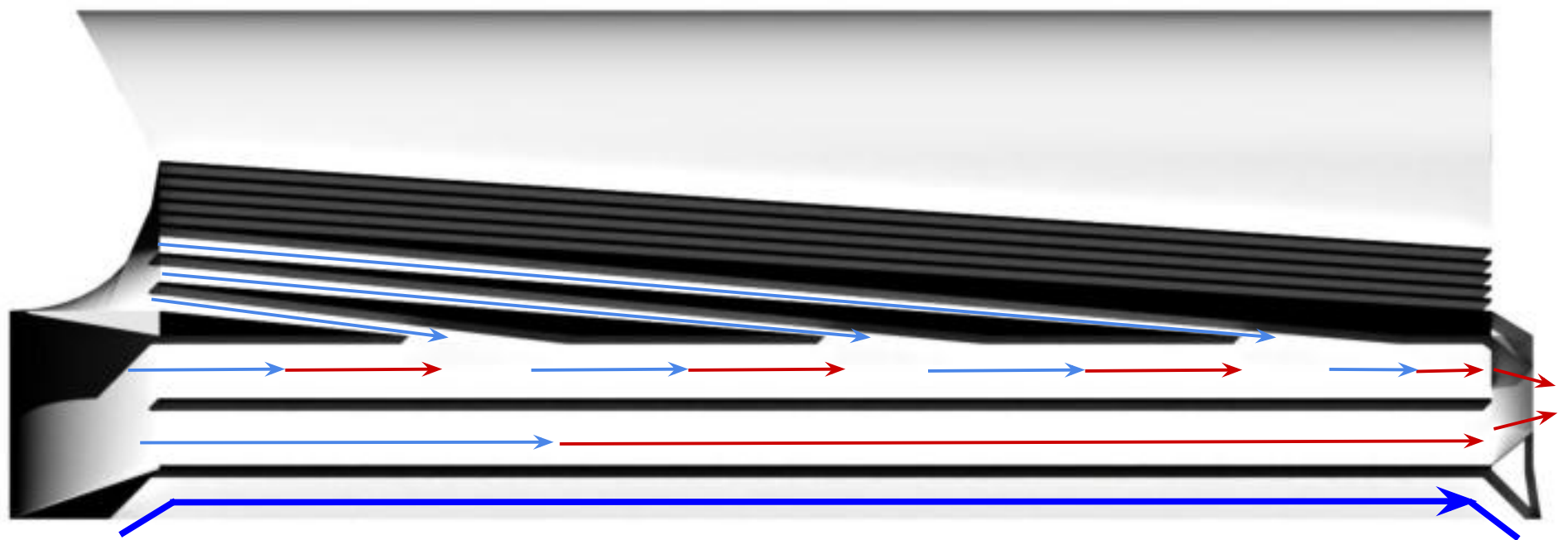
The Technology

Model V11 -Section Diagram-



The Technology

Model V11 -Section Diagram-

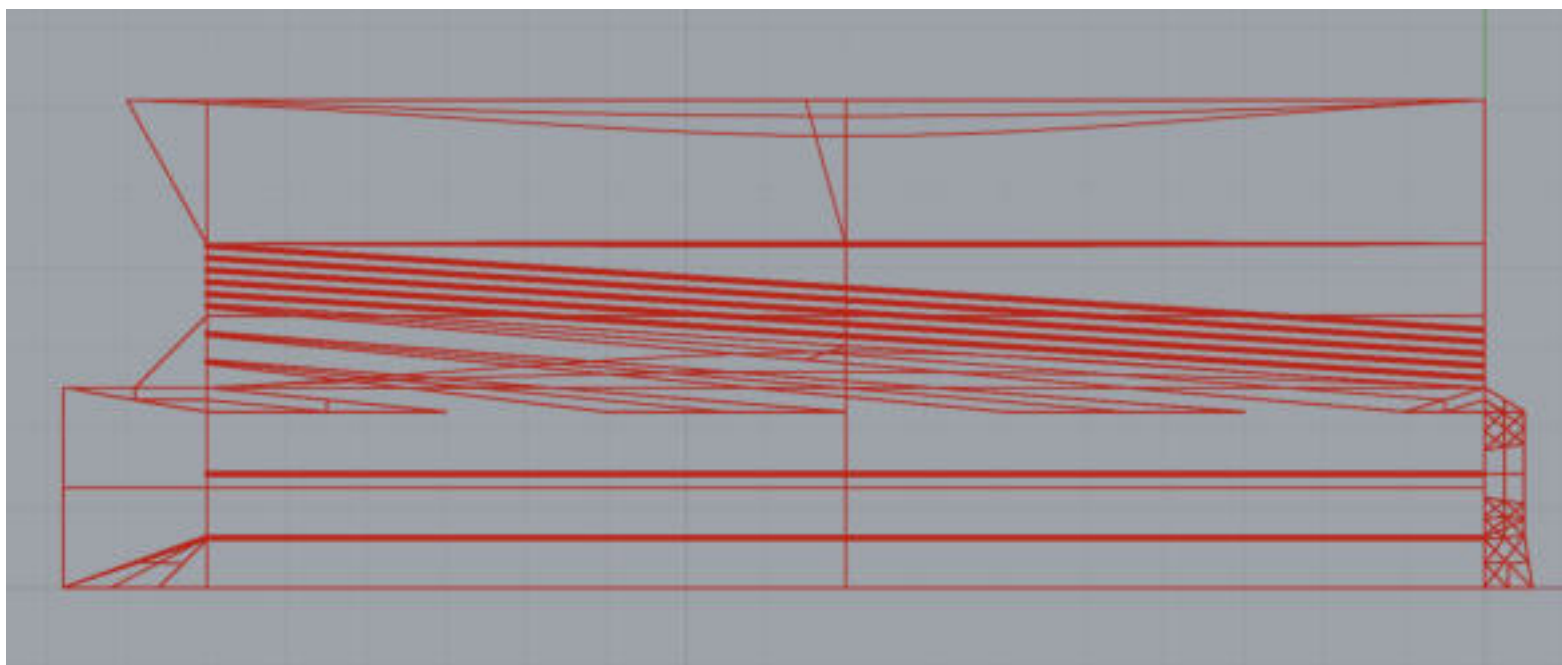
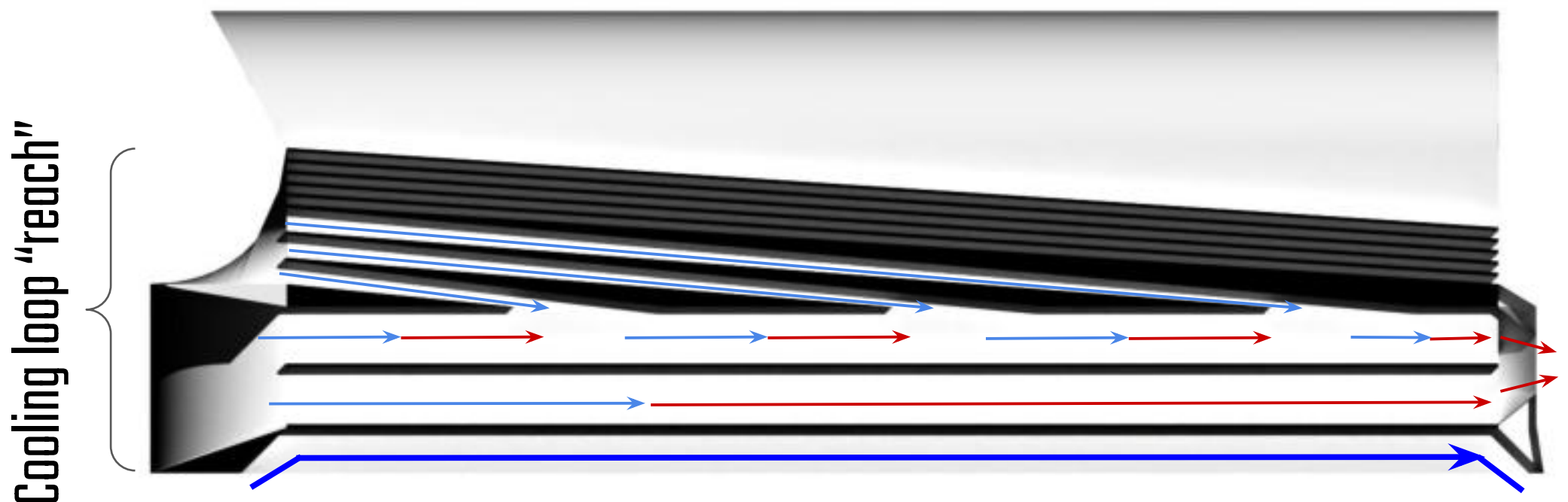


- water cooling circulation pipes
- hot (warmed) air
- cold (inflow) air

The Technology

Model V11 -Section Diagram-

Enclosed individual server cooling loop
—visualized earlier— extends to from the
floor to the top of the “cooling fins”



- water cooling circulation pipes
- hot (warmed) air
- cold (inflow) air

The Hybrid Cooling Data Center

Structure Measurements

General Specifications:

422.73 meters²

200 Servers (120x80cm each)

192 meters² of server space

Each Module:

1,830cm long

360cm wide

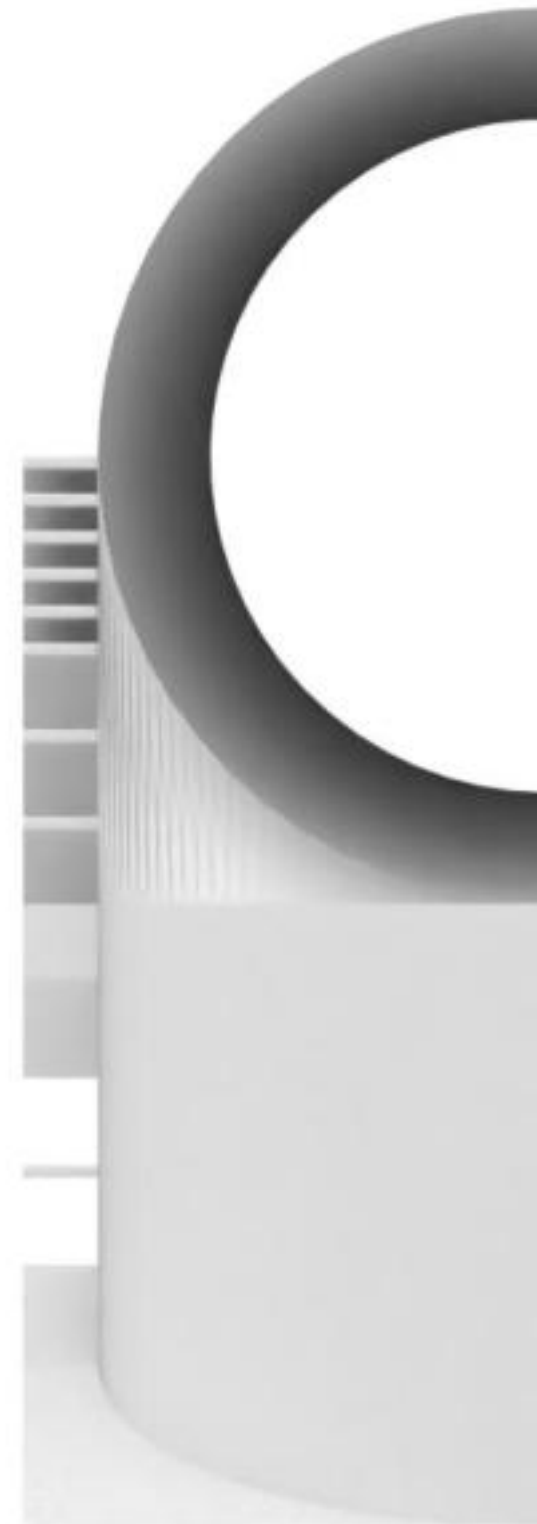
610cm tall (~250cm without generator fan & auxiliary vents)

Other Measurements:

30cm wide “Cooling Corridor” (between each module)

200 to 220cm interior height (depending on server height)

With normal rentable service/hosting data centers, usually only one third to one fourth ($\frac{1}{3}$ to $\frac{1}{4}$) of the building is utilized as server space (about half of the space if hallways are included)



The Hybrid Cooling Data Center

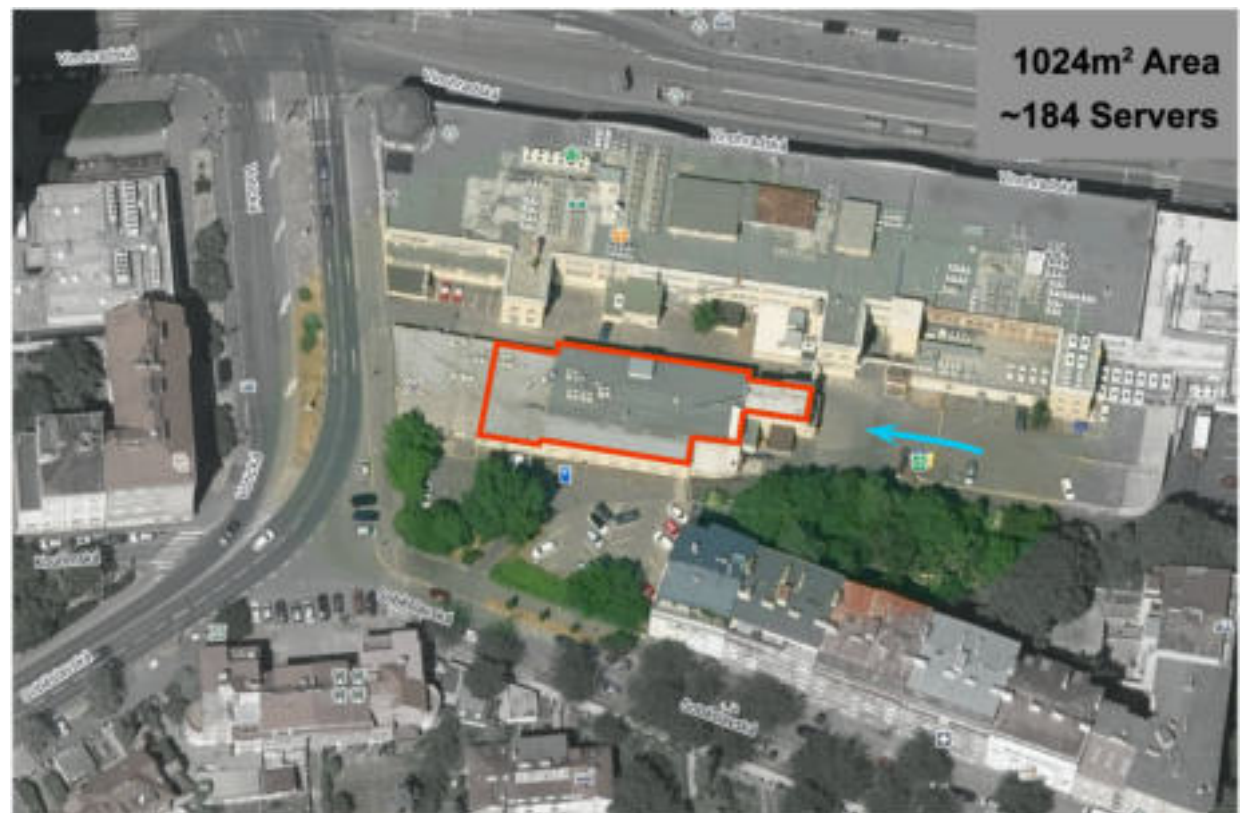
Data Center Comparison

With normal rentable service/ hosting data centers, usually only one third to one fourth of the building is utilized as server space (about half of the space if hallways are included)

This is the “COOLHOUSING Data Center”

According to typical spatial distributions of these kinds of data centers, it holds approximately 184 servers (server racks);

Similar capacity to this proposal, however; at 200 server capacity, my design takes up the area of 425 meters²– less than half of that facility, while also potentially being many times more efficient.



The COOLHOUSING Data Center (for comparison)

The Hybrid Cooling Data Center

Data Center Comparison

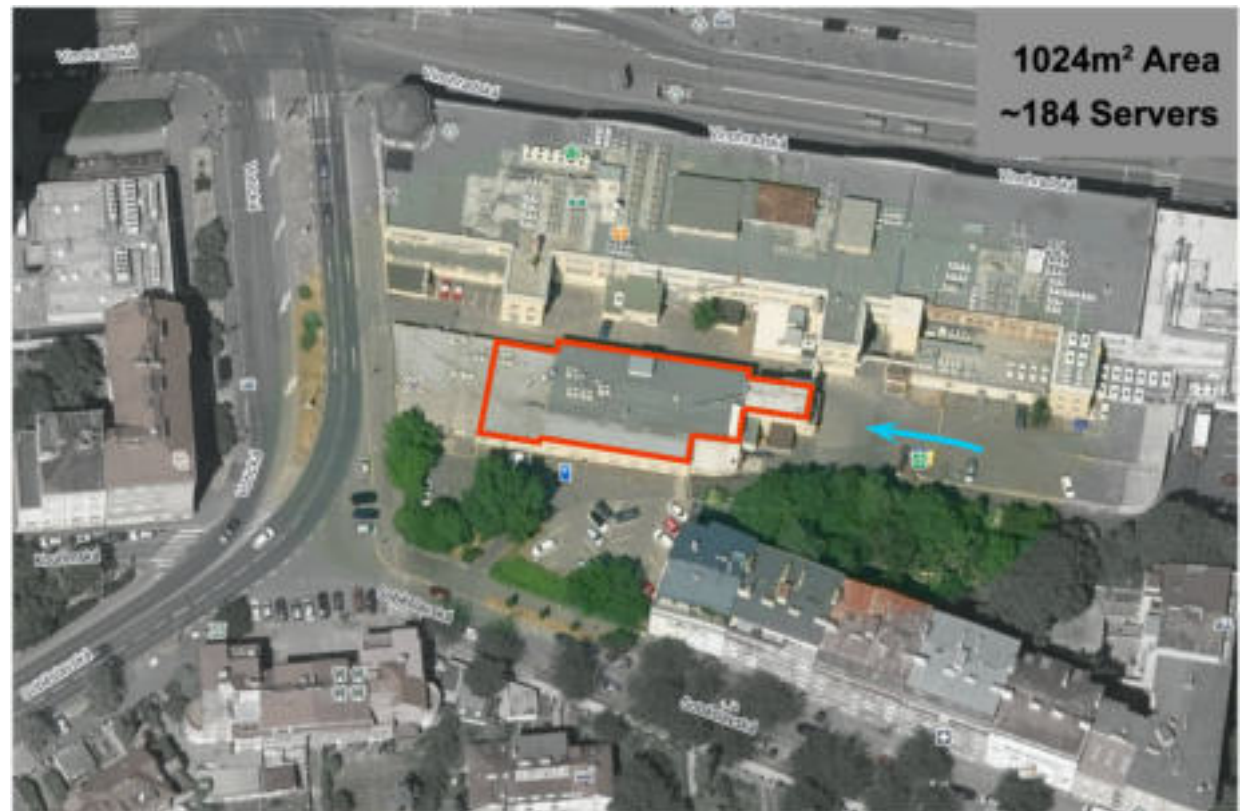
With normal rentable service/ hosting data centers, usually only one third to one fourth of the building is utilized as server space (about half of the space if hallways are included)

The “COOLHOUSING Data Center”

- Approximately 184 servers
(server racks)
- 1024 meters²

My Design (200 server model):

- 200 server capacity
(slightly better capacity)
- 425 meters² (less than half area used)
- Potentially/likely more efficient cooling



The COOLHOUSING Data Center (for comparison)

The Hybrid Cooling Data Center

Data Center Material Structure

Structure includes: Copper exterior shell, with a reinforced (structural) steel rib support structure, and concrete/stone base.

The Shell:

-

Copper (5mm): 3,300 kg

Copper (5mm) GWP: 9,500 kg CO₂ eq/m³

-

Copper (10mm): 6,600 kg

Copper (10mm) GWP: 19,000 kg CO₂ eq/m³

-

(All statistics rounded up for any potential mistakes during construction)

The Hybrid Cooling Data Center

Data Center Material Structure

Structure includes: Copper exterior shell, with a reinforced (structural) steel rib support structure, and concrete/stone base.

The Ribs:

-

Structural Steel: 6.9 m^3 (per rib)

Structural Steel: 69 m^3 (per module) (x6 for the basic structure = 414 m^3)

Structural Steel: 541,650 kg (if solid)

Structural Steel GWP: 373,000 kg CO₂ eq/ m^3 (if solid)

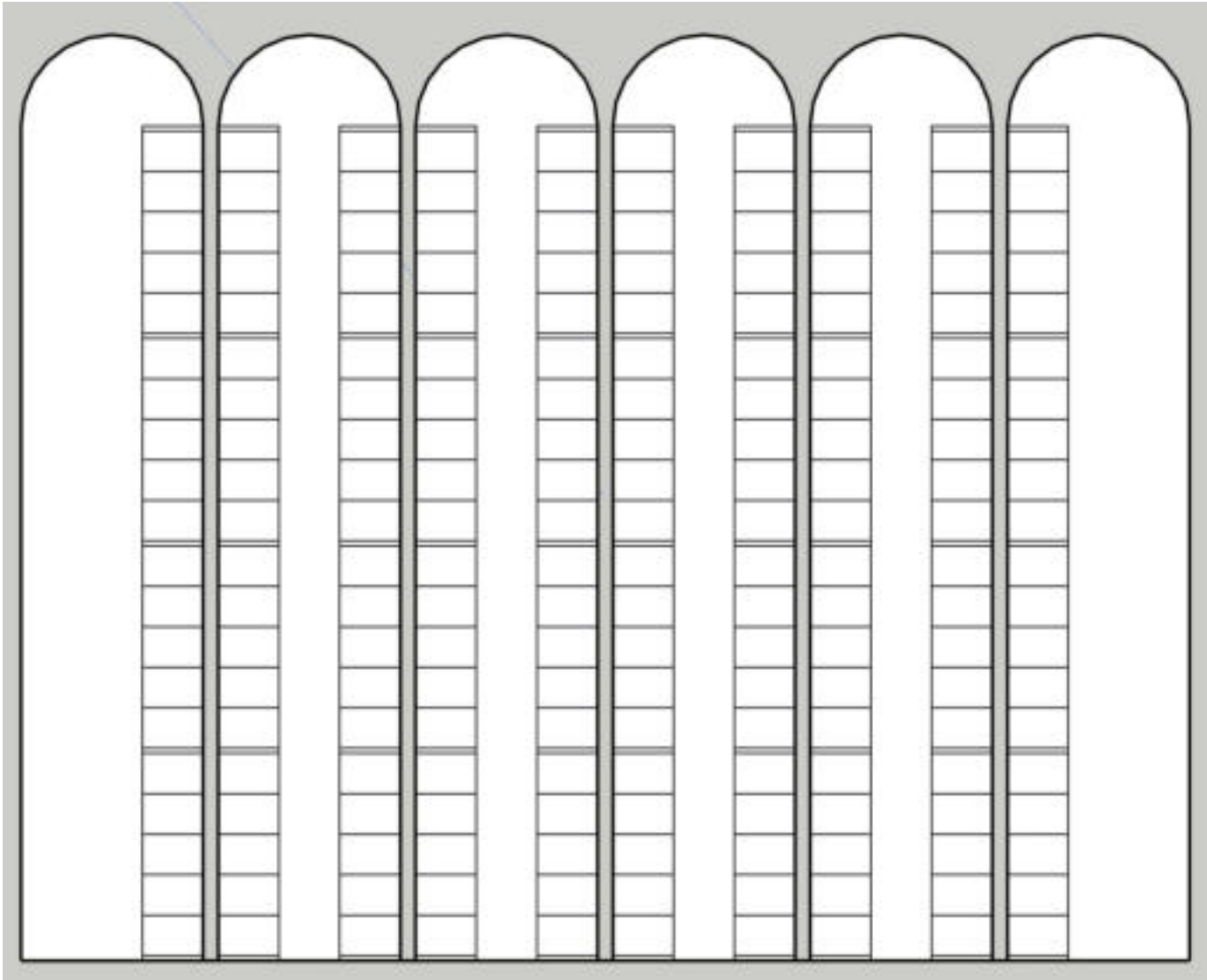
-

(All statistics rounded up for any potential mistakes during construction)

The Hybrid Cooling Data Center

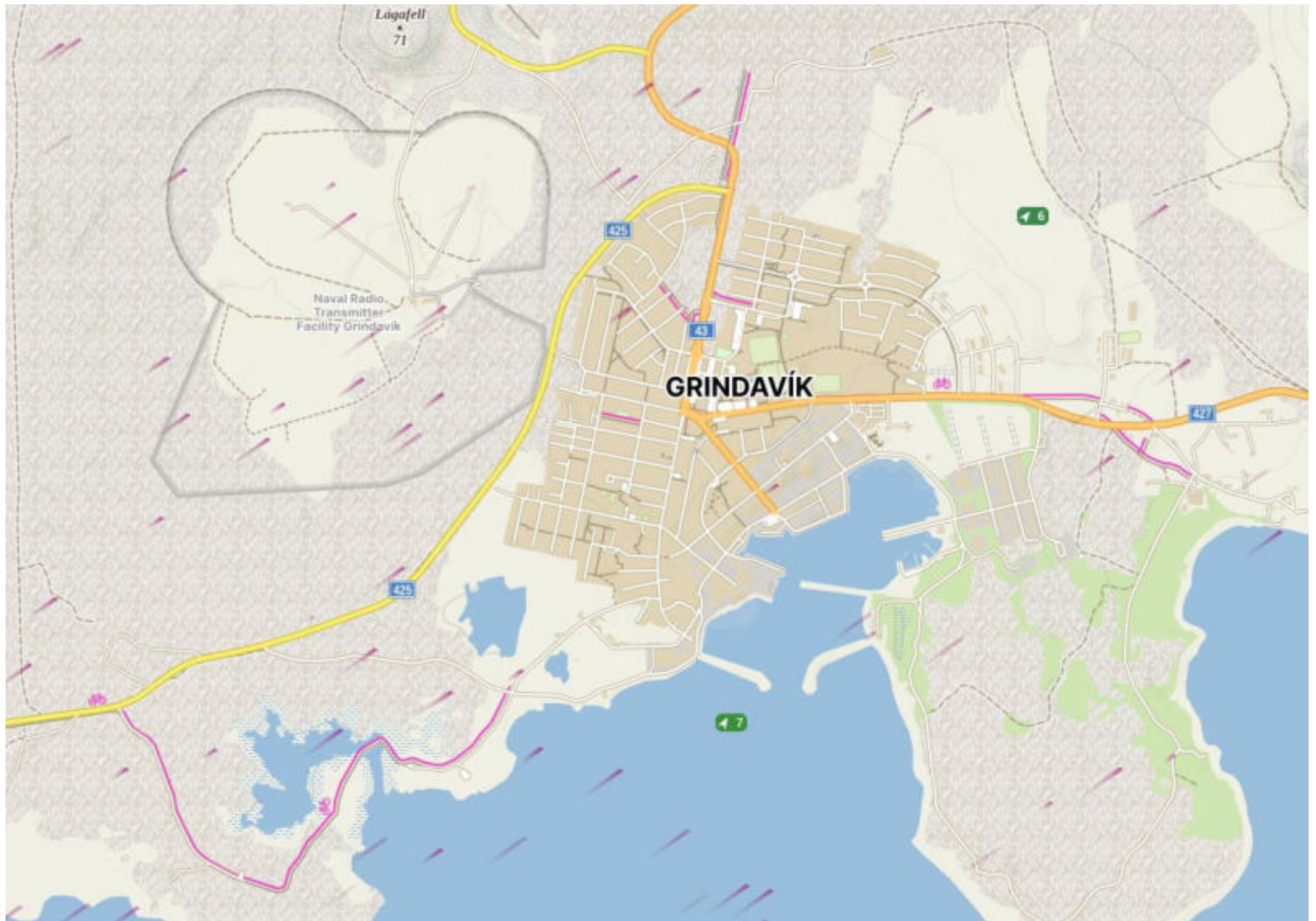
Data Center Structure

Between the roof fan and the servers, in the ceiling, four high capacity battery packs (400x340*) would sit, each taking up 13.6 m² of horizontal area (with a 20 to 40 cm height maximum)

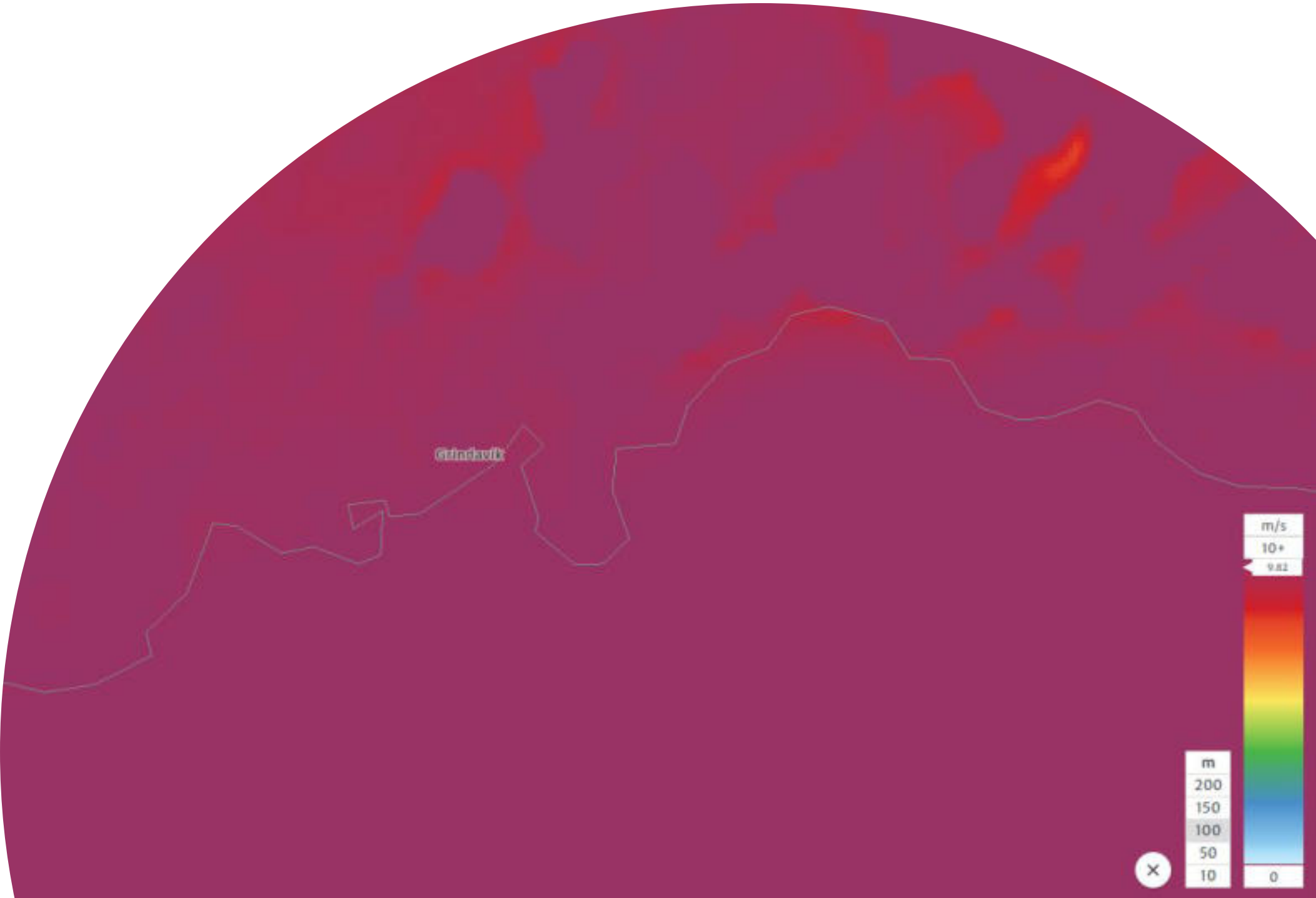


*(400x340cm—could be interchanged for two smaller batteries; 360x200 or 180x400)

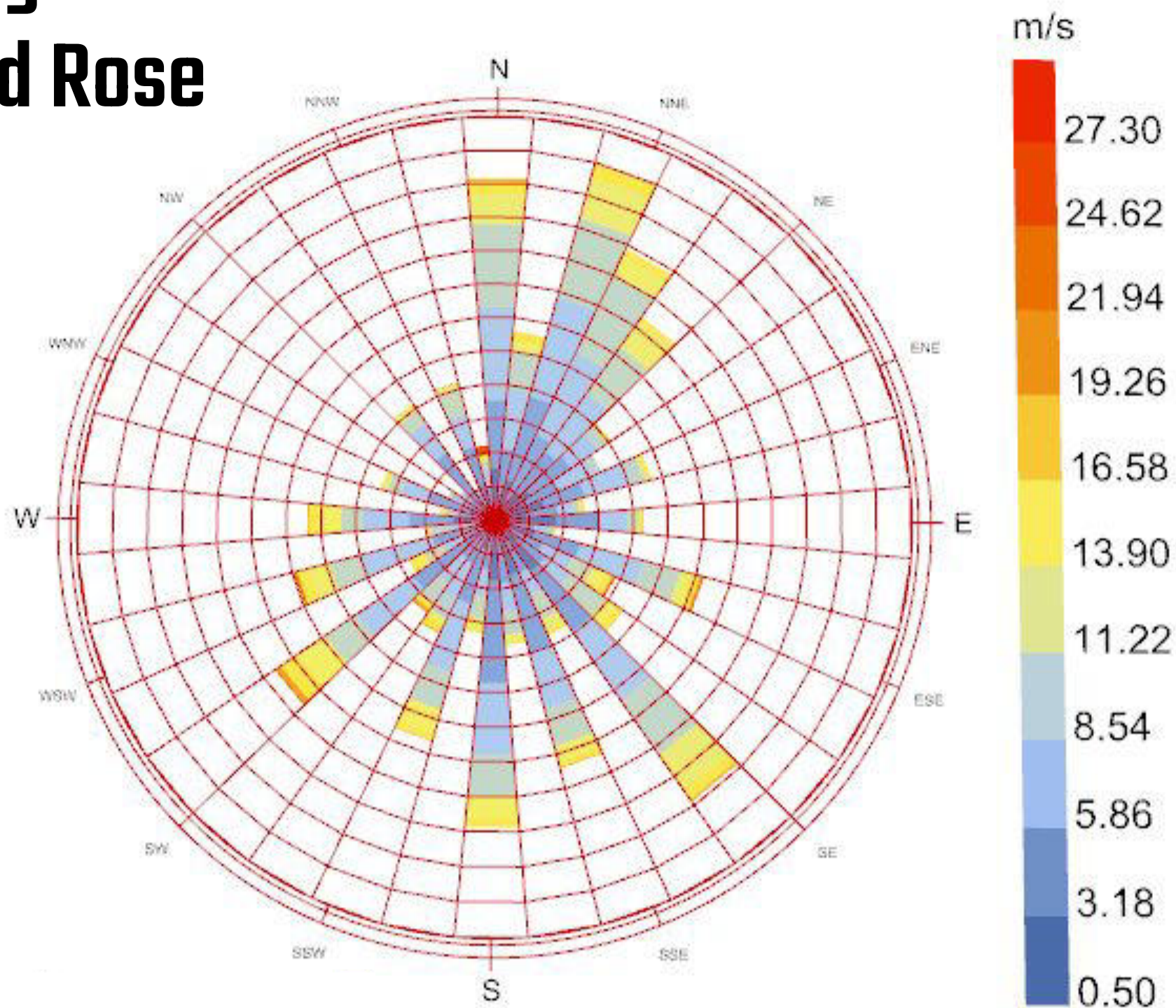
Location



Location



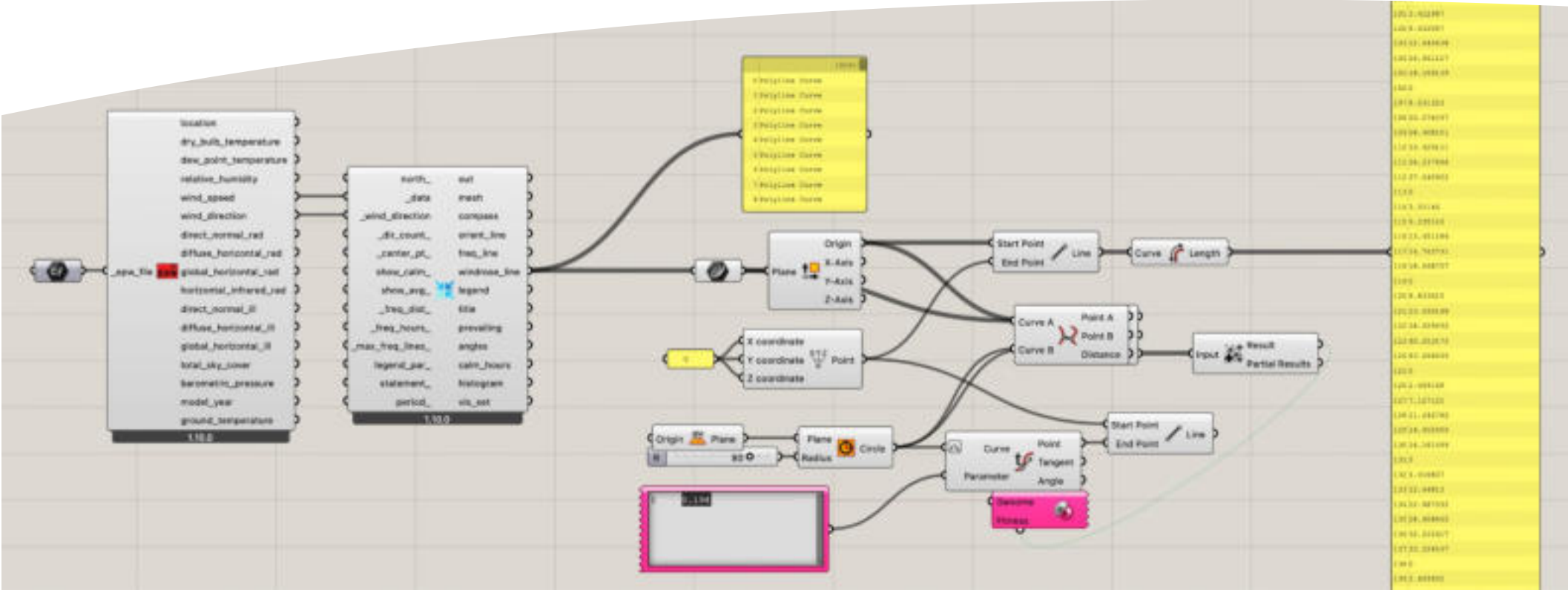
Ladybug LD Wind Rose



Wind Speed (m/s)
time-zone: 0.0
source: SRC-TMYx
city: Keflavik.Intl.AP
country: ISL
period: 1/1 to 12/31 between 0 and 23 @1
Calm for 1.75% of the time = 153 hours.
Each closed polyline shows frequency of 0.6% = 50 hours.

Ladybug

LD Wind Rose



KPIs



The amount of water used for the purposes of cooling, of similar capacity. Unit: Liters of water used over a unit of time.

The amount of electricity saved through this design compared to other popular strategies, of similar capacity. Unit: Average amount of processing versus watts used.

The area of free space free of the structure, compared to the average of similar capacity. Unit: m² and % of site area.

The amount of human time spent in or around the structure, compared to structures of similar capacity. Unit: Time.

Thank You for Your Time

And Please Have A Nice Day